

HRMS Project Report Summary

Course: DBMS Lab| **Stack:** PHP 8.x, MySQL 8.0+ (3NF), Bootstrap 5

1. Project Overview

The Human Resource Management System (HRMS) Portal is a 3-tier web application built to digitize workforce management. It handles employees, departments, and attendance tracking while enforcing 3rd Normal Form (3NF) relational integrity and Role-Based Access Control (RBAC).

2. Tech Stack & Architecture

- **Database:** MySQL 8.0+ (3NF Normalized Schema, InnoDB Engine)
- **Backend:** PHP 8.x (MySQLi Prepared Statements, Session Authentication, PRG Pattern)
- **Frontend:** HTML5, CSS3, Bootstrap 5 (Solid Dark Theme, English/Bengali Support)
- **Security:** BCrypt Password Hashing, Role-Based Access Controls

3. Database Schema (3NF Summary)

Table	Primary / Foreign Keys	Key Columns & Description
users	id (PK)	name, email (Unique), password (Bcrypt), role (Admin/Dept/Staff), is_deleted
departments	id (PK)	dept_name (Unique), is_deleted

Table	Primary / Foreign Keys	Key Columns & Description
employees	id (PK), dept_id (FK -> departments.id)	first_name, last_name, email, salary, hire_date, is_deleted
attendance_logs	id (PK), emp_id (FK -> employees.id)	log_date, log_time, status (Present/Absent/Leave), is_deleted

4. Key System Features

- **Soft Delete & Recovery System:** Records are soft-deleted (`is_deleted = 1`) to preserve relational integrity, with instant recovery options and permanent deletion for Admins.
- **Role-Based Access Control (RBAC):**
 - *Admin* (`admin@hrms.com`): Full system CRUD, user setup, restore/purge privileges.
 - *Manager* (`dept@hrms.com`): Department & employee soft-deletion controls.
 - *Staff* (`staff@hrms.com`): Read-only view of attendance logs.
- **Bilingual UI & Dark Theme:** Toggle between English and Bengali across key system modules.

5. Project Structure

```
hrms-project/
├── config/db.php      # Database setup & connection
├── includes/          # Header, navbar, and footer layout
├── index.php          # Dashboard analytics
├── login.php / logout.php # Auth logic & quick-fill testing
├── users.php          # Admin account management
├── departments.php    # Department CRUD & soft-delete
├── employees.php      # Employee directory
└── attendance.php     # Daily timekeeping logger
```